PICARD GROUPS DO NOT GROW UNDER EXTENSION OF THE BASE FIELD WHEN THE GLOBAL FUNCTIONS ARE SEMILOCAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof's first sentence performs the special-case reduction to the delivered companion; the remaining sentence disposes of the 'e.g.' clause.

Allowed standard prerequisites (editorial).

- Change of rings and Picard groups. delivered companion lemma-change-rings-pic. The framework lemma of which this is the field-extension special case.
- Finite-dimensional algebras are Artinian with finitely many maximal ideals. Algebra, Lemmas algebra-lemma-finite-dimensional-algebra and algebra-lemma-artinian-finite-nr-max. The e.g. clause.

Source reference. Original lemma 0CDY: sources/stacks-chow/tags.txt, source lines 15558–15558. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00J6: sources/stacks-chow/tags.txt, source lines 704–704. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00J7: sources/stacks-chow/tags.txt, source lines 705–705. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k'/k be a field extension. Let X be a scheme over k such that*

(1) *X is quasi-compact and quasi-separated, and*
(2) *$R = H^0(X, \mathcal{O}_X)$ is semi-local, e.g., if $\dim_k R < \infty$.*

Then the pullback map $\mathrm{Pic}(X) \to \mathrm{Pic}(X_{k'})$ is injective.

Proof. Special case of Lemma 0CDY. If $\dim_k R < \infty$, then R is Artinian and hence semi-local (Algebra, Lemmas 00J6 and 00J7). □

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